Deep Neural Network Based Anomaly Detection For Smart Surveillance System

Shubhangi D C
Dept. of Computer Science and Engineering,
VTU CPGS, Kalaburagi
Karnataka, India
drshubhangipatil1972@gmail.com

Mohammed Abdul Waheed
Dept. of Computer Science and Engineering,
VTU CPGS, Kalaburagi,
Karnataka, India
dr.mawaheed@gmail.com

Baswaraj Gadgay
VTU Regional Office, Kalaburagi,
Karnataka, India
baswaraj_gadgay@vtu.ac.in

Vishwanath P
Dept. of Electronics and Communication Engineering H. K. E. Society's S. L. N. College of Engineering, Raichur
Karnataka, India
vishalpetli73@gmail.com

Varsha Tadalapur
Student, Dept. of Computer Science and Engineering,
VTU CPGS, Kalaburagi, Karnataka, India
varshatadalapur@gmail.com

2022 IEEE North Karnataka Subsection Flagship International Conference (NKCon) | 978-1-6654-5342-4/22/$31.00 ©2022 IEEE | DOI: 10.1109/NKCON56289.2022.10126698

***Abstract*— Searching for anomalies in surveillance film is a practise that is becoming more and more popular. Anomaly detection is the procedure of spotting irregularities in data. When we talk about "abnormal data," we mean data that drastically differs from its typical behaviour. Fraud detection, malfunction detection, and intrusion detection are just a few of the uses for anomaly detection. You Only Look Once is a method for quickly recognising objects (YOLO). It is an object identification system that is capable of quickly locating objects in images, real-time coverage, and video streams. An instant object recognition system is the YOLO Convolutional Neural Network (CNN). CNNs are classifier-based systems that can identify patterns in images and interpreting them as collections of well-organized data. Both the intersection over union (IOU) and mean average precision (mAP) values of YOLOv3 are quick and exact. This detection method is significantly slower than other detection methods that get comparable results. This study suggests utilising a modified YOLOv3 algorithm to track things automatically and notify people if anything seems off. The performance of the algorithm was also contrasted with that of other algorithms, including CNN and decision trees. Our system continuously records live video while abnormal things like fire and weapons are identified. High detection precision and quick processing are features of our technology.**

Keywords— CNN (Convolution Neural Network), Decision tree, intersection over union (IOU) values, Mean average precision (mAP), OpenCV, YOLOv3 (You Only Look Once Version3)

I. INTRODUCTION

Video surveillance is a useful tool in a range of industries, including law enforcement, transportation, environmental monitoring, and others, to boost security and public safety. It currently plays a key role in crime prevention, traffic control and investigation. Detecting unusual events such as traffic accidents and traffic violations, crime and natural disasters is one of the most important, difficult, and time-sensitive tasks in automated video surveillance. As a result, current research has focused a lot of attention on video anomaly detection. Finding data occurrences that depart from nominal trends is a broad, important, and challenging topic of academic study termed "anomaly detection". It can be utilised for a variety of purposes, such as spacecraft tracking, hardware security, cyber security, and medical security. Given the important role that video anomaly detection can play in maintaining security and, in certain circumstances, preventing major tragedies, the ability to make real-time decisions is one of the most important outcomes of a video anomaly detection system. Real-time detection of aberrant events may help with immediate countermeasures in the case of remote robberies, fires, and traffic accidents. The method of identifying an event in a video is known as video event detection. A subtype of video event detection is video anomaly detection. A significant amount of data is produced and shown during the video anomaly detection process. High demand exists for software that can track movement and record it on camera. There are three categories of video anomaly detection techniques: manual, semi-automatic, and completely automated. A unique individual is appointed to watch live footage using the manual anomaly detection technique. In any case, it is challenging to keep track of the data gathered using this technique. Since running the software for several days will fill up the hard disk, continuous recording is not possible. The hiring of a third party is not necessary for a semi-automated operation. Fully automated means the same as semi-automated in that the abnormal video is not examined by a video analyser. On the other hand, the machine examines the video and detects an anomaly in the movement that triggers the alarm. Object recognition is one of the most challenging problems in image processing. Although there are other methods of object identification, in this essay we will focus on YOLOv3 (you only look once). You Only Look Once Version 3 (YOLOv3) is a real-time object recognition system that can detect specific things in pictures, live streams, and movies. To find an item, YOLO employs features learned from a deep convolutional neural network. Ali Farhadi and Joseph Redmon were responsible for the first three iterations of the YOLO phrase. The third version of YOLO, which is the topic of this article, was released in 2018, two years after the first version, which was released in 2016. A more advanced version of YOLOv2 and YOLO is YOLOv3. OpenCV Deep Learning or Kera's. The size of the prediction map is the same as the feature map above because the YOLO prediction uses 11 convolutions.

978-1-6654-9790-9/22/$31.00 ©2022 IEEE

A system for quickly and accurately recognising objects is the YOLO Convolutional Neural Network (CNN). CNNs are classifier-based systems that can identify patterns in incoming images and translate them into organised arrays of data (view image below). The advantage of YOLO is that it is faster than other networks while maintaining accuracy. When tested, the complete image is examined, enabling the model to draw conclusions about the image's broader context. cfg file: The third version of the YOLO system is called YOLOv3 (YOLOv3 Paper). The yolov3.cfg file stores the pre-trained weights and model architecture of the neural network. The 80 object classes are listed in a file named coco that the model can recognise. Names. Weights file: A weights file with a speed training regimen. The folder for the dark net contains this file. Also found in the data/directory is the coco. Names file, which contains a list of objects the model can identify. The cfg/directory contains a configuration data file called coco. Data.

II. Related Work

T. Celik et. al. [1], has proposed a real-time fire detect or that combines advanced object information with fire colored pixel statistics. In the first stage of this technique, background noise is eliminated, and preceding objects are found that are primarily the result of an object's movement or transitory changes to its position in the picture. When the front pixels are recognized by colors like fire, the second step is initiated. The primary result of this procedure is the removal of the front, inflammable material. [2] Girshick Ross, Donahue Jeff, Darrell Trevor, Mali k Jitendra. 2016 Performance for object detection on the canonical PASCAL VOC Challenge datasets peaked in the latter years of the competition. The most effective strategies made use of complex ensemble systems, which frequently blended numerous low-level graphics with high-level information. In this work, we present a simple and scalable detection approach that improves mean average precision (mAP) by more than 50% over the previous best VOC 2012 result (mAP of 62.4%). Our strategy is based on two ideas: (1) supervised pre training for a supplementary task, followed by domain specific fine-tuning, significantly improves performance when labelled training data is scarce; and (2) when labelled training data is scarce, combining numerous low-level visuals with high-level context produced the best results. Because it mixes CNN and region proposals, the final model is known as R-CNN or region-based convolutional network. [3] Lucio Marcenaro et. al. raised the issue of premature fire detection and smoke based on color factors and movement analysis. Early fire noticing is divided into three stages. a) The classification of pixels with fire or smoke. b) Segmenting by region. c) Examination of the regions that qualify. As a result of this effort, pixels in fire/smoke photos are categorized, clustered into regions, and the internal motion of areas is examined. [4] S. Nadon et. al. proposed fire detection algorithm to monitor fires using AVHRR images. It takes two procedures to identify a fire: marking potential fires and putting out fake ones. Important knowledge on fire operations is made available by the findings across the nation. [5]J. Redmon et. al. Introduced YOLO, a new discovery method. The previous acquisition function is altering the dividers to access. Through their outputs, they are able to combine all aspects of object detection into a single neural network. The statistical fire colour model is composed of three laws. According to the first rule, an RGB pixel's value for the red component must be greater than the average value for the red components over the entire image. The second rule states that a pixel's red value must be higher than its green value, which must then be higher than its blue value. The final rule considers the relative amounts of the colours Red, Blue, and Green. The prior rules are supplemented by all of these rules. Error is caused by non-linearities in the fixed camera, sudden changes in lighting, as well as by some items burning with different fire colours. However, this strategy fails when there is only smoke and no red pixels. [6]M. Trinath et al. propose an IOT based solution for the problem. They use temperature and smoke sensors in their system. The sensors are expensive, fragile, and susceptible to damage from many natural forces, which is the system's major flaw. [7]Grega et al. proposed a method for automatic detection of dangerous situations in CCTV systems, through the use of image processing and machine learning. Using sliding window approaches, fuzzy classifiers, and canny detectors, weapons and knives were found in videos. The authors made their built-in dataset and detection method available.

III. Proposed Method

In our suggested system, the YOLOv3 algorithm is used to detect weapons using the following technique: A dataset with three different sorts of weapons—handguns, knives, and big guns—and different images of fire, which were downloaded from Google, is initially constructed. This dataset was trained for weapon classification and images of fire using the YOLOv3 (You Only Look Once) algorithm. After the data has been trained, the system is able to recognize the type of weapon being used and unpredicted fire in real-time video input from security cameras, as well as the confidence score for each weapon. An alarm will sound if any anomalies are detected. The authorities will be contacted if the weapon is found.

*YOLOv3:*YOLOv3 is a real-time tracking tool that locates specific elements in movies, live streams, and images. To locate something, it makes use of learning attributes. YOLO is a Convolutional Neural Network (CNN) that does real-time object identification. YOLO is faster than other networks, and it also has superior accuracy.YOLOv3 is a superior version of YOLO and YOLOv2. Deep learning libraries such as Keras or OpenCV are used to implement YOLO.

YOLO v3 object detection procedure:

- The inputs consist of several form images (m, 416, 416, 3).
- This image was received from YOLO v3 to a Convolutional Neural Network (CNN).
- The output volume of the aforementioned calculation is after flattening the remaining two dimensions (19, 19, 425).
- In this instance, a grid of 19 x 19 results in 425 results per cell.
- Considering that a grid has five anchor boxes, 425 is equal to 5 * 85.

- 85 = 5 + 80, where 5 is equal to (pc, bx, by, bh, and bw) and 80 is the total number of classes to be identified.
- The output is a list of the classes and bounding boxes that have been selected. There is a six-digit representation for each bounding box (pc, bx, by, bh, bw, c). Each bounding box can be represented by 85 different values if we expand c to an 80-dimensional vector.
- Finally, we employ the IoU (Intersection over Union) and Non-Max Suppression to prevent choosing overlapping boxes.

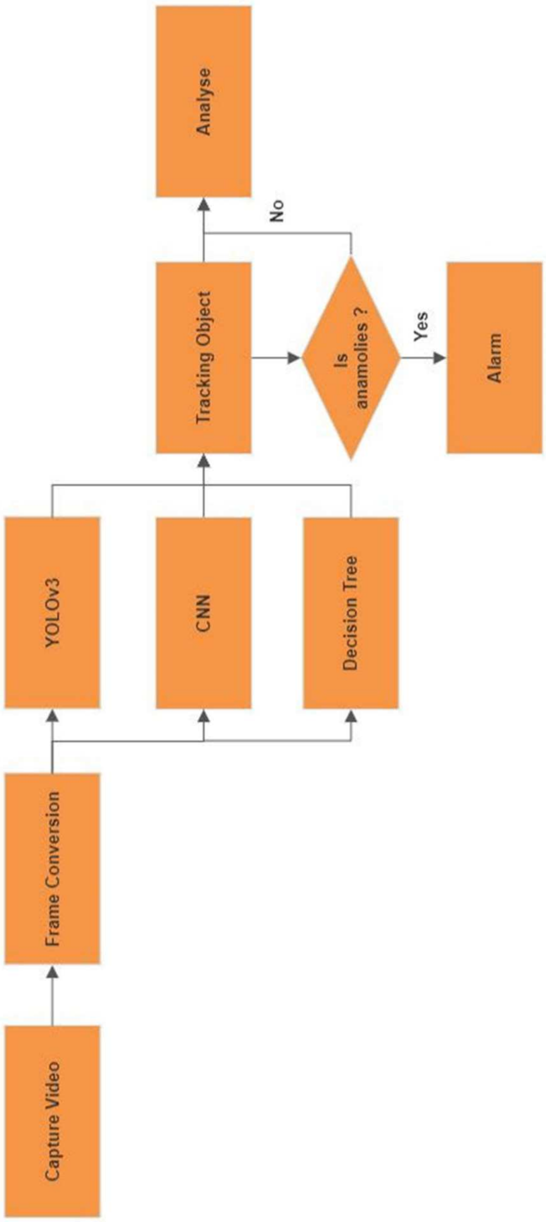

Fig.1: System Architecture

A. Capture Video

OpenCV offers a straightforward user interface for taking live video with the camera (webcam). A grayscale version of the video is shown. A video capture object must be made to record video. A device index or the name of a video file could be included. The camera can be uniquely identified by its device index, which is a number. By supplying a 0 or 1, we can select the camera. After that, each frame of the video may be seen.

B. Frame Conversion

Frames are the process of extracting still images from a video. The video is frame-by-frame transformed, then saved in the photos folder.

C. Tracking using CNN

The CNN architecture is a well-liked neural network design for computer vision problems. The fact that CNN automatically deletes features from images implies that the network discovers important elements.

Convolutional, pooling, and fully linked layers are the three basic subsystems of CNN. 1024 nodes would be present in a multi-layer approach for a grayscale image with the dimensions of 32*32 pixels. Spatial positions in the image are lost during flattening. Internal feature representation is learned using tiny squares of input data to maintain the spatial relationship between picture elements.

i. The convolutional layer: a convolutional layer, which includes feature maps and filters. Filter processors can only do one layer of processing. There is no way to change these filters. A feature map is the end product after beginning with pixel values. The result is a feature map that uses just one filter layer. The filter is applied to the image one pixel at a time. When only a small number of neurons are active, a feature map is produced.

ii. A pooling layer: To reduce dimensionality, a pooling layer is employed. Following one or two convolutional layers, pooling layers are added to generalize features seen in earlier feature maps. As a result, there is a lower possibility that you will be over fit throughout the training session.

iii. Fully connected layer: After gathering, combining, and pooling the data from the convolutional layer, the fully connected layer is used to assign features to class likelihood. These layers use soft-max or linear activation functions.

D. Tracking Using YOLOV3

Versions 1 and 2 of YOLO employ SoftMax techniques to convert scores into probabilities. When two things are mutually exclusive, this method is suitable. YOLOv3 employs an approach of numerous labels. To ascertain the probability that input would fall under a specific label, an independent logistic classifier is used. Loss is determined by computing the binary-cross entropy of each label. To simplify things, the SoftMax function is not used. To predict whether an object will exist, bounding box optimization in YOLO v3 employs logistic regression.

An anchor box that overlaps the ground truth box the most gives an item the highest object-ness score of 1. The anchor box is vacant when the overlap exceeds the predetermined level. There is one anchor box mapped for every ground truth box. If the anchor box is not chosen and assigned to the bounding box, the confidence loss will take the place of the classification and localization losses.

E. Decision Tree

A decision tree is a graphic representation of each possible resolution to a problem based on a set of criteria. To separate all the label training items, we attempt to establish conditions on the features at each level or node of a classification decision tree. The cost of gathering and comparing features, their ability to distinguish between objects in the domain, and the stability or separability of classes of objects are all things to consider while using the dataset in its simplest form. To enable incremental, local tree updates, an intelligent subset of extractable attributes is selected to accommodate new items or instances of previously misclassified objects. A decision tree is built to determine a collection of features that are all important during the development process. An image is initially divided into a grid via the YOLOv3 algorithm. Each grid cell expects a specific number of border boxes (sometimes called anchor boxes) to surround objects that perform well in the predefined classes. Each bounding box identifies a single object, and it includes a confidence value that indicates how accurate the forecast should be. The most common sizes and shapes are discovered by combining the original dataset's ground truth boxes, which are then used to build the border boxes.

F. Models

All three designs—YOLOV3, CNN, and Decision Tree—are used by the system. It keeps track of the object and informs the user via text and speech whenever an anomaly is found.

IV. RESULT ANALYSIS

A real-time object detection system finds anomalies like fire and weapons from webcams or surveillance videos. It detects the anomalies and sends an alert message to the concerned authorities by sending messages. Our project helps for public security purposes. When fire or weapons are found, the system will collect the evidence of the scenario, an alarm will sound to alert the public and send a message and current location to the crime branch to take the situation under control.

Table 1: Comparison of YOLO with Different Datasets

S .No	Model	Dataset	Accuracy
1	YOLOV2	Weapon Dataset	0.7845
2	YOLOV3	Weapon Dataset	0.8623
3	YOLOV2	COCO Dataset	0.8943
4	YOLOV3	COCO Dataset	0.9231

Fig.2: Main

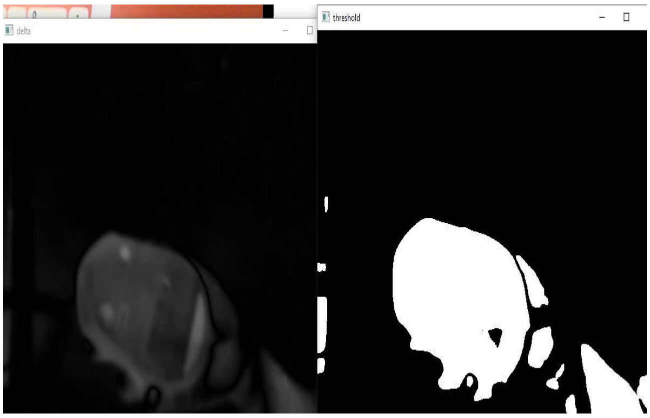

Fig.3: Motion Detection

Fig.4: Gun detection

Fig.5: Fire detection

Fig.6: Gun Detection

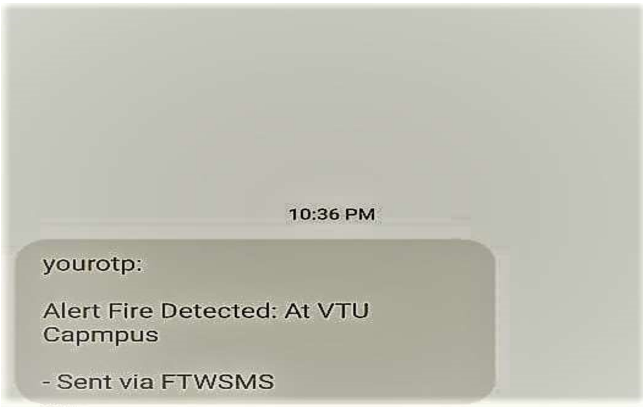

Fig.7: Alert message for concerned authorities

V. CONCLUSION

YOLOV3 is a better object detection method, therefore, accurately tracking the object with a 91 percent accuracy and a 51 percent loss. Using YOLOv3, several items in COCO datasets may be differentiated. Performance metrics for YOLOv3 are compiled based on the photo classes analyzed. The precession value of the class increases as mAP increases. For tracking several objects in traffic surveillance film, YOLOv3 and OpenCV are employed. Multiple objects are recognized and tracked in various movie frames. Utilizing YOLOV3 has several advantages, but its remarkable speed—which enables it to analyze 45 frames per second—is by far the most important. Additionally, the idea of generic object representation is understood by YOLOV3. Comparable to CNN, this object detection technique is effective. The long-term objective of the proposed system is to classify and increase the number of weapon types. The detection accuracy of the weapon can be increased by using different algorithms. Finding a concealed weapon that is invisible to the normal camera is one method to make this work better. It is also feasible to analyze people's behavior to find any questionable activity, such as hiding a weapon, and improve this monitoring system. The alarm mechanism can be modified to alert numerous users if the weapon is found. These qualities of a surveillance system can aid in deterring violent crimes and ensuring public safety.

REFERENCES

[1]. Girshick Ross, Donahue Jeff, Darrell Trevor, Malik Jitendra. Region Based Convolutional Networks for Accurate Object Detection and Segmentation. [J]. IEEE transactions on pattern analysis and machine intelligence,2016,38(1).
[2]. Molchanov V, Vishnyakov B V, Vizilter Y V, et al. Pedestrian detection in video surveillance using fully convolutional YOLO neural network[C]//Automated Visual Inspection and Machine Vision II. International Society for Optics and Photonics, 2017, 10334: 103340Q.
[3]. J. Redmon and A. Farhadi, “YOLO9000: better, faster, stronger,” in 2017 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017, Honolulu, HI, USA, July 21-26, 2017, pp. 6517–6525, 2017.
[4]. Hao Ye, Geoffrey Ye Li, Biing-Hwang Juang. Power of Deep Learning for Channel Estimation and Signal Detection in OFDM Systems. [J]. IEEE Wireless Commun. Letters,2018,7(1).
[5]. W. Anser, M Fraz, M Shahzad , , ”Two Stream Deep CNN-RNN Attentive Pooling Architecture for Video-based Person Re-identification”, Proceedings of the 23rd Iberoamerican Congress on Pattern Recognition, Nov, 2018, Madrid , Spain.
[6]. Wangpeng He, Zhe Huang, Zhifei Wei, Cheng Li, Baolong Guo. TFYOLO: An Improved Incremental Network for Real-Time Object Detection[J]. Applied Sciences,2019,9(16).
[7]. R M S Bashir, M Shahzad, M M Fraz , ”VR-PROUD: Vehicle Reidentification using PROgressive Unsupervised Deep architecture”, Pattern Recognition, Vol. 90 , No. 1, PP. 52-65, Jan, 2019.
[8]. Tian Y, Yang G, Wang Z, et al. Apple detection during different growth stages in orchards using the improved YOLO-V3 model[J]. Computers and electronics in agriculture, 2019, 157: 417-426.
[9]. Luo J, Wu X, Luo Y, et al. Real-world image datasets for federated learning[J]. arXiv preprint arXiv:1910.11089, 2019.
[10]. Z. Zhao, P. Zheng, S. Xu and X. Wu, ”Object Detection With Deep Learning: A Review,” in IEEE Transactions on Neural Networks and Learning Systems, vol. 30, no. 11, pp. 3212-3232, Nov. 2019, doi: 10.1109/TNNLS.2018.2876865.
[11]. Choi J, Chun D, Kim H, et al. Gaussian yolov3: An accurate and fast object detector using localization uncertainty for autonomous driving[C]//Proceedings of the IEEE International Conference on Computer Vision. 2019: 502-511.
[12]. Yang, Hongmin Deng. GC-YOLOv3: You Only Look Once with Global Context Block[J]. Electronics,2020,9(8).
[13]. Haodong Pan, Jue Jiang, Guangfeng Chen. TDFSSD: Top-Down Feature Fusion Single Shot MultiBox Detector[J]. Signal Processing: Image Communication,2020,89.
[14]. S. Ahmed, A. Ahmed, I. Mansoor, F. Junejo, and A. Saeed, “Output feedback adaptive fractional-order super-twisting sliding mode control of robotic manipulator,” Iranian Journal of Science and Technology, Transactions of Electrical Engineering, vol. 45, no. 1, pp. 335–347, 2021.
[15]. S. Ahmed, H. Wang, and Y. Tian, “Adaptive high-order terminal sliding mode control based on time delay estimation for the robotic manipulators with backlash hysteresis,” IEEE Transactions on Systems, Man, and Cybernetics: Systems, vol. 51, no. 2, pp. 1128–1137, 2021.

[16]. SKJ Rizwi, MA Azad, M.M. Fraz , ”Spectrum of Advancements and Developments in Multidisciplinary Domains for Generative Adversarial Networks (GANs)”, Archives of Computational Methods in Engineering, Vol. 2021 , No. 1, Apr, 2021.